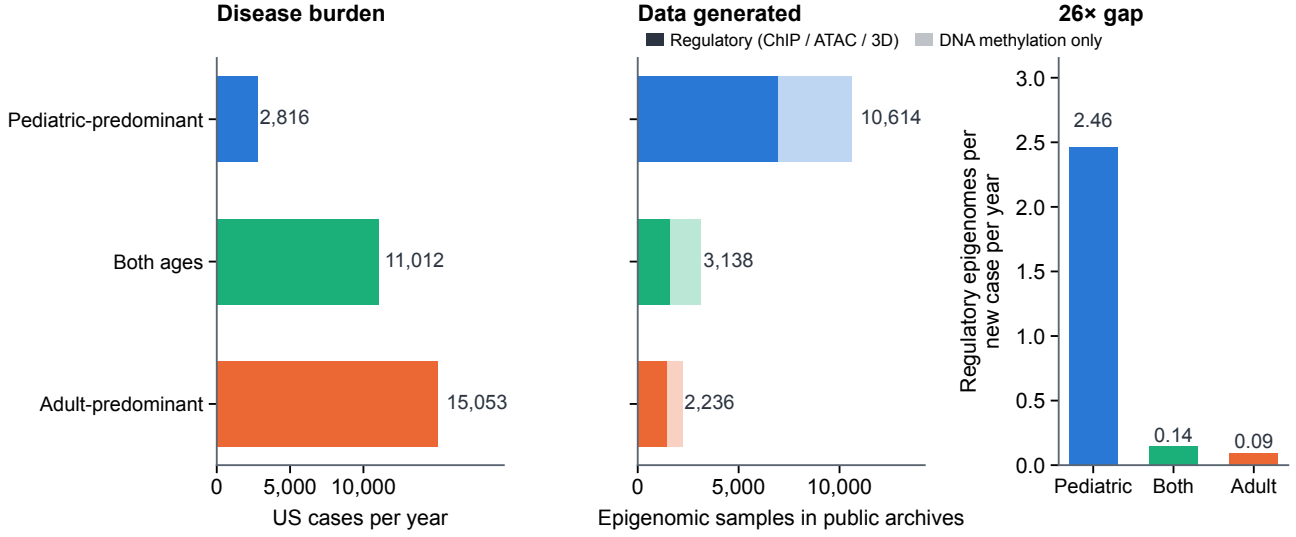
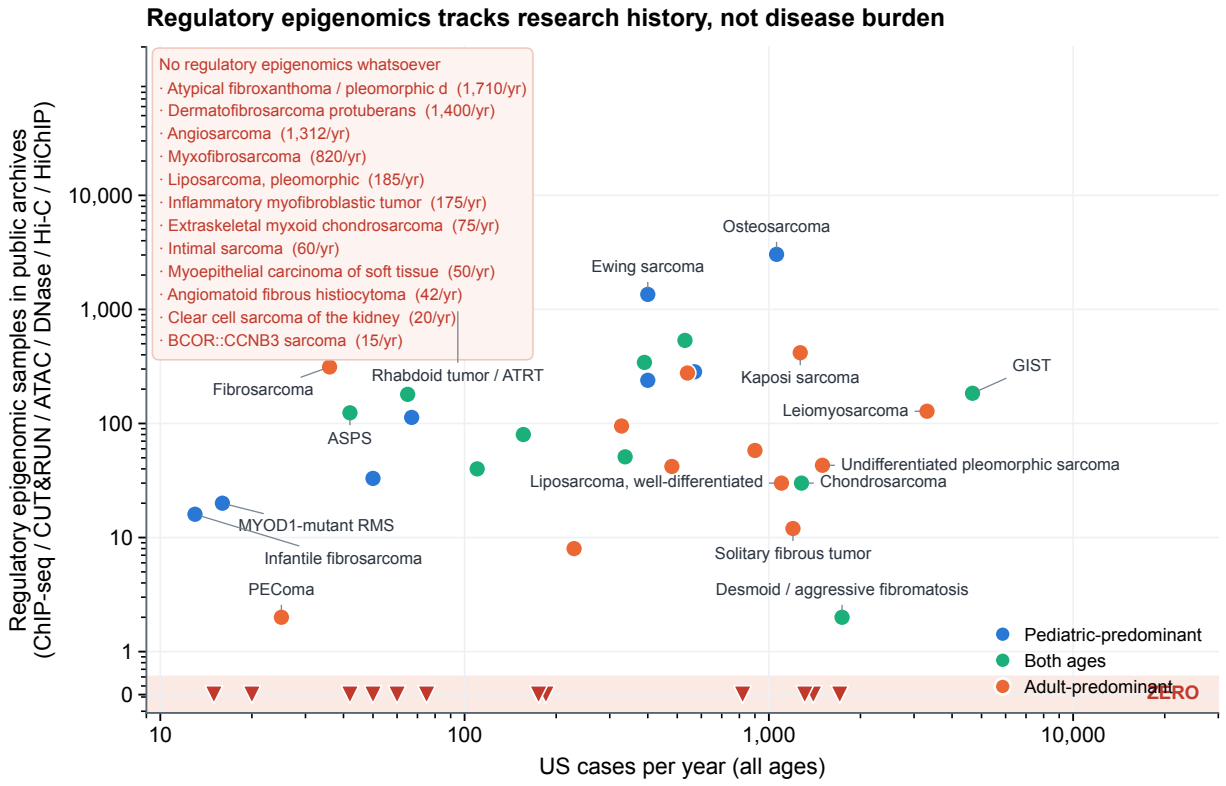
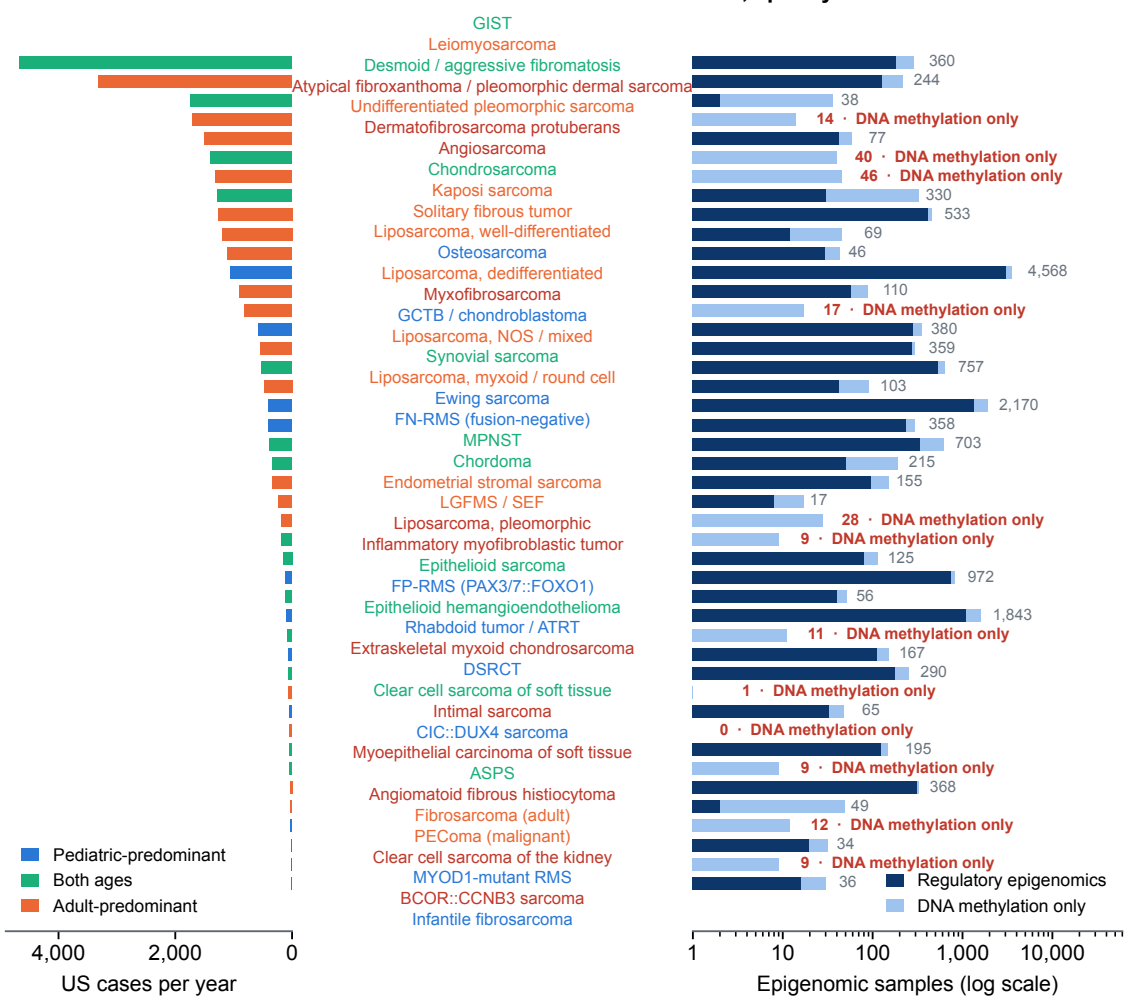
Adult sarcomas carry ~5× the disease burden and a small fraction of the epigenomic data

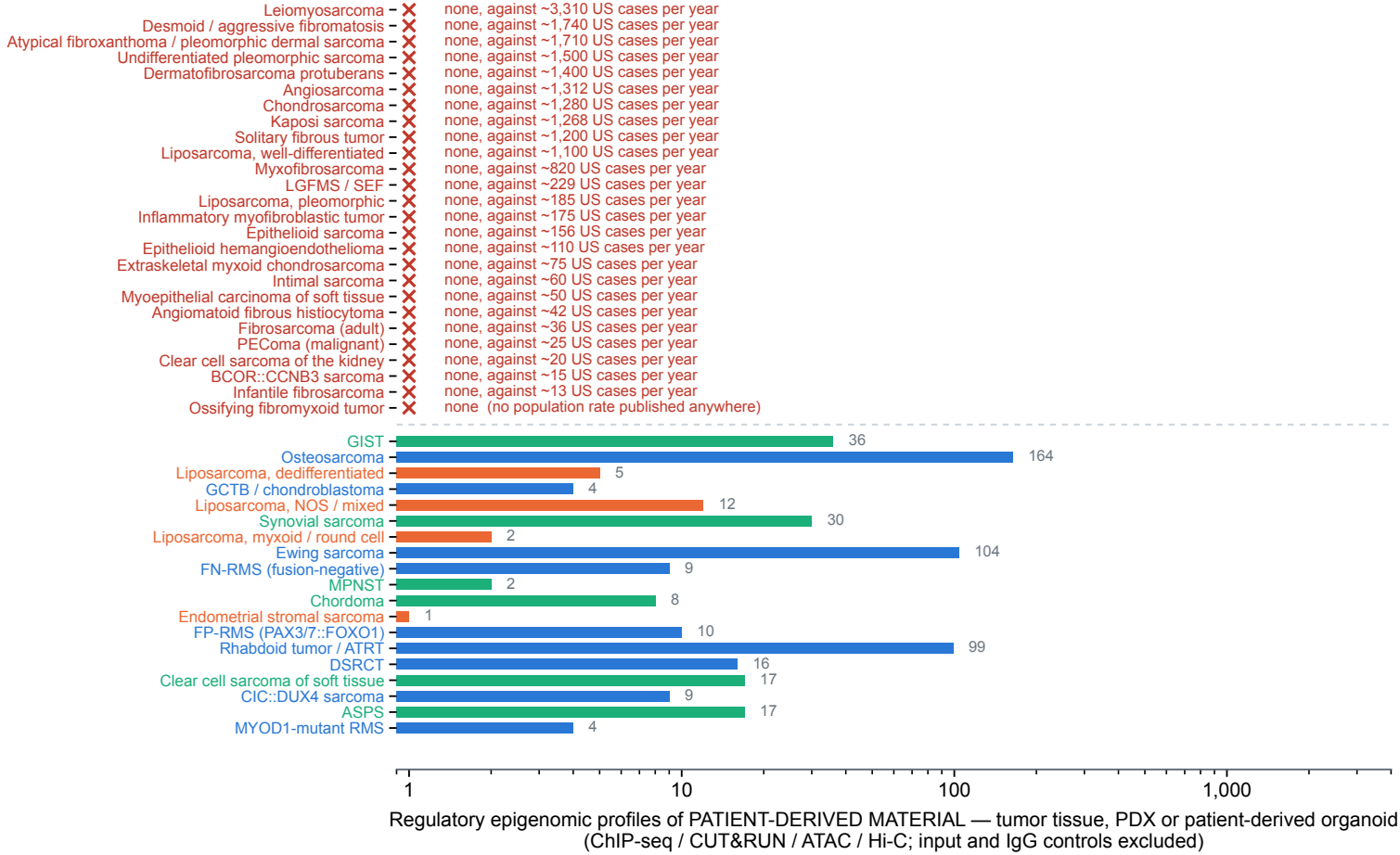



Regulatory epigenomics tracks research history, not disease burden
No regulatory epigenomics whatsoever
· Atypical fibroxanthoma / pleomorphic d (1,710/yr)
· Dermatofibrosarcoma protuberans (1,400/yr)
· Angiosarcoma (1,312/yr)
· Myxofibrosarcoma (820/yr)
· Liposarcoma, pleomorphic (185/yr)
· Inflammatory myofibroblastic tumor (175/yr)
· Extraskeletal myxoid chondrosarcoma (75/yr)
· Intimal sarcoma (60/yr)
· Myoepithelial carcinoma of soft tissue (50/yr)
· Angiomatoid fibrous histiocytoma (42/yr)
· Clear cell sarcoma of the kidney (20/yr)
· BCOR::CCNB3 sarcoma (15/yr)
Regulatory epigenomic samples in public archives
(ChIP-seq / CUT&RUN / ATAC / DNase / Hi-C / HiChIP)
10,000
1,000
100
10
1
0
Osteosarcoma
Ewing sarcoma
GIST
Kaposi sarcoma
Fibrosarcoma
Rhabdoid tumor / ATRT
Leiomyosarcoma
ASPS
Undifferentiated pleomorphic sarcoma
Liposarcoma, well-differentiated
Chondrosarcoma
MYOD1-mutant RMS
Infantile fibrosarcoma
Solitary fibrous tumor
PEComa
Desmoid / aggressive fibromatosis
Pediatric-predominant
Both ages
Adult-predominant
ZERO
10
100
1,000
10,000
US cases per year (all ages)

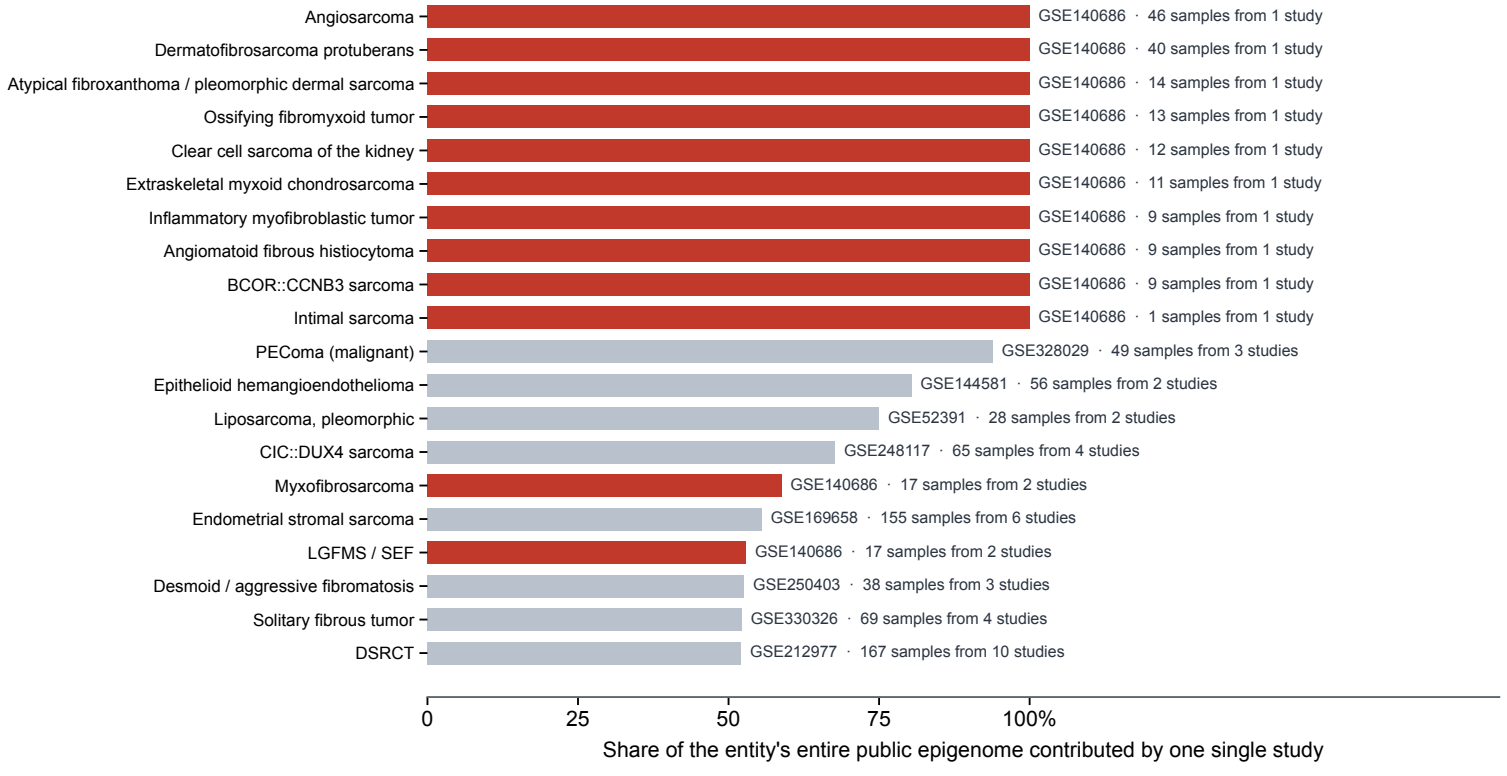
Nine sarcoma entities with a published incidence rate — two of them above 1,300 US cases a year — have DNA methylation profiling and nothing else



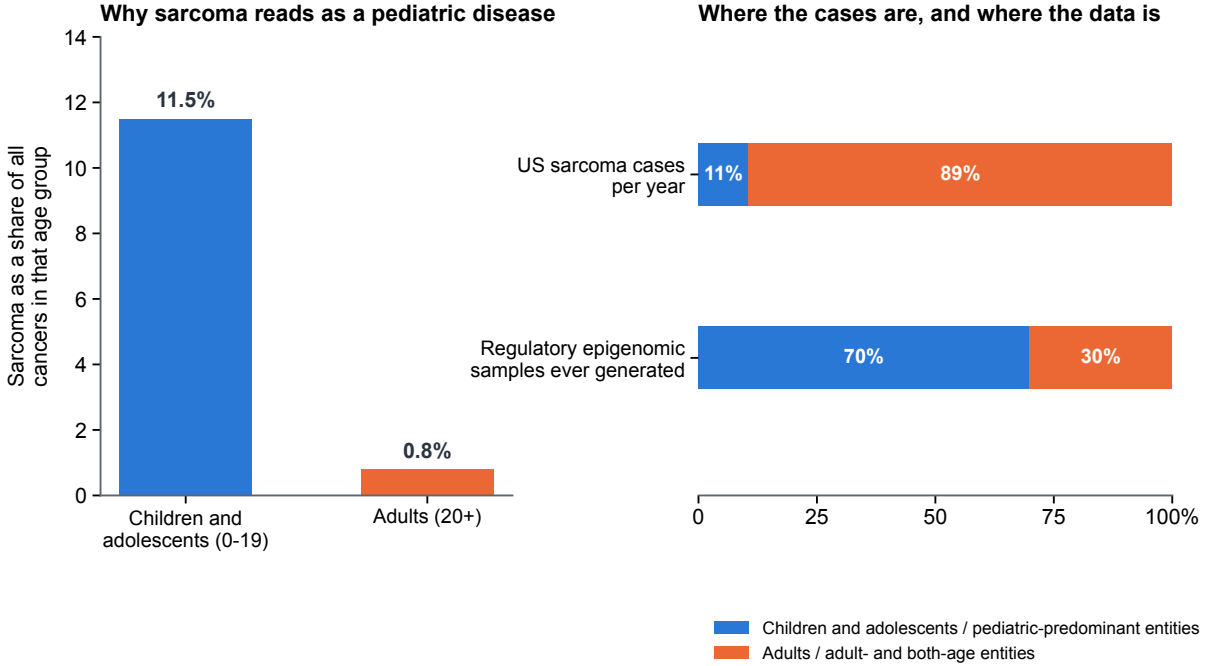
26 of 45 sarcoma entities have never had a single patient's tumor profiled for active chromatin, chromatin accessibility or 3D genome architecture



Most sarcoma entities rest on one deposit — and for many of them that deposit carries no entity label in GEO at all

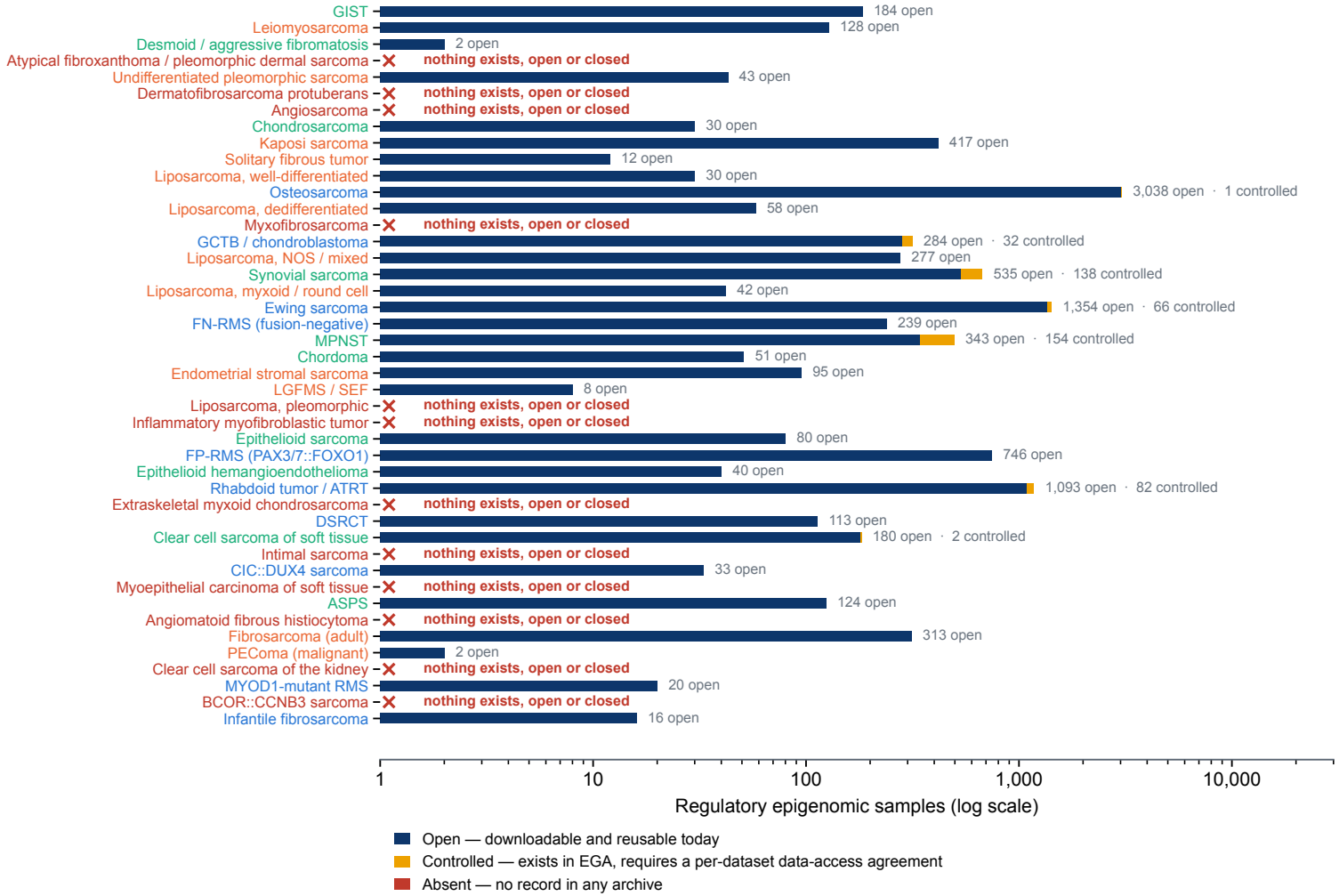


Sarcoma is 1% of adult cancer and 11% of childhood cancer — and the epigenomic literature followed the 11%



Sources: sarcoma share of all cancers and US case counts, Siegel et al., Cancer Statistics 2026 (PMID 41528114; 4,110 bone + 13,910 soft tissue = 18,020) and SEER 21 age distribution; pediatric share of childhood malignancy, Siegel DA et al., JNCI 2023 (PMID 37433078; ICCC groups VIII + IX = 11.5%). Sample counts from this atlas (T13).

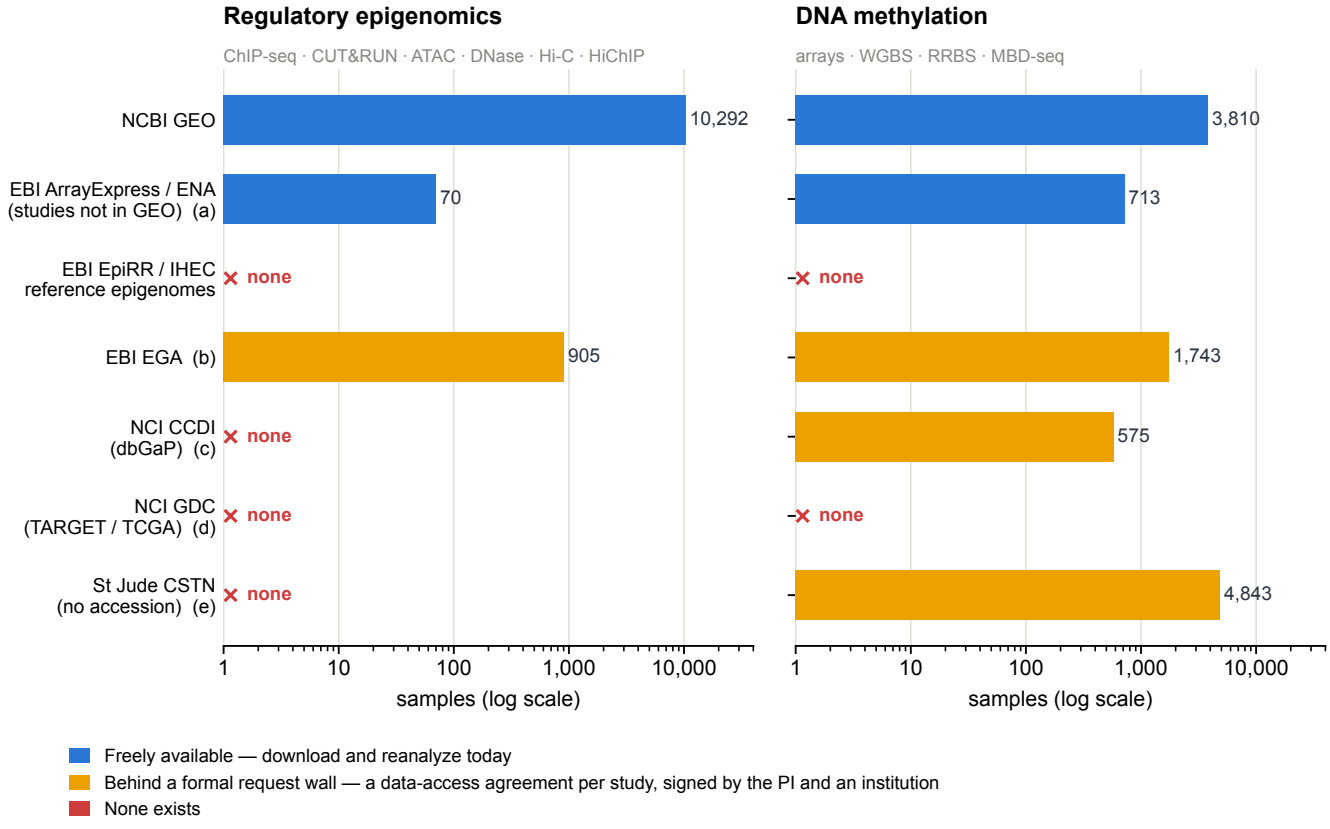
Some of the gap is unmeasured biology; some of it is data that exists and cannot be reused
1,207 regulatory epigenomes sit behind data-access committees — but not one covers the nine entities with nothing at all



A further 430 regulatory epigenomes sit in EGA under controlled access in cohorts whose subtype EGA does not state — RMS (any subtype): 413; Sarcoma, unspecified / mixed cohort: 17.
The largest is the St Jude CSTN rhabdomyosarcoma ChIP-seq (EGAD00001004312 + EGAD00001006398), none of which is in GEO. They are excluded from the bars rather than attributed by guesswork.

Regulatory epigenomics for sarcoma is mostly open. DNA methylation is mostly not.

92% of the 11,267 regulatory samples can be downloaded today, against only 39% of the 11,684 methylation samples.



Unit: samples, except CCDI, which resolves to participants. Duplicates removed; ChIP input and IgG controls excluded from the regulatory counts.

(a) The 23 studies that exist only at EBI, sized from the ENA and BioStudies APIs. E-MTAB-9875 (1,974 assays) is held out — it is the sarcoma methylation classifier, the same material as GSE140686 in GEO.

(b) Includes the St Jude CSTN deposit (469 samples), which is counted here rather than separately because EGA is where it lives. (c) CCDI has no ChIP-seq in its assay vocabulary at all, for any disease.

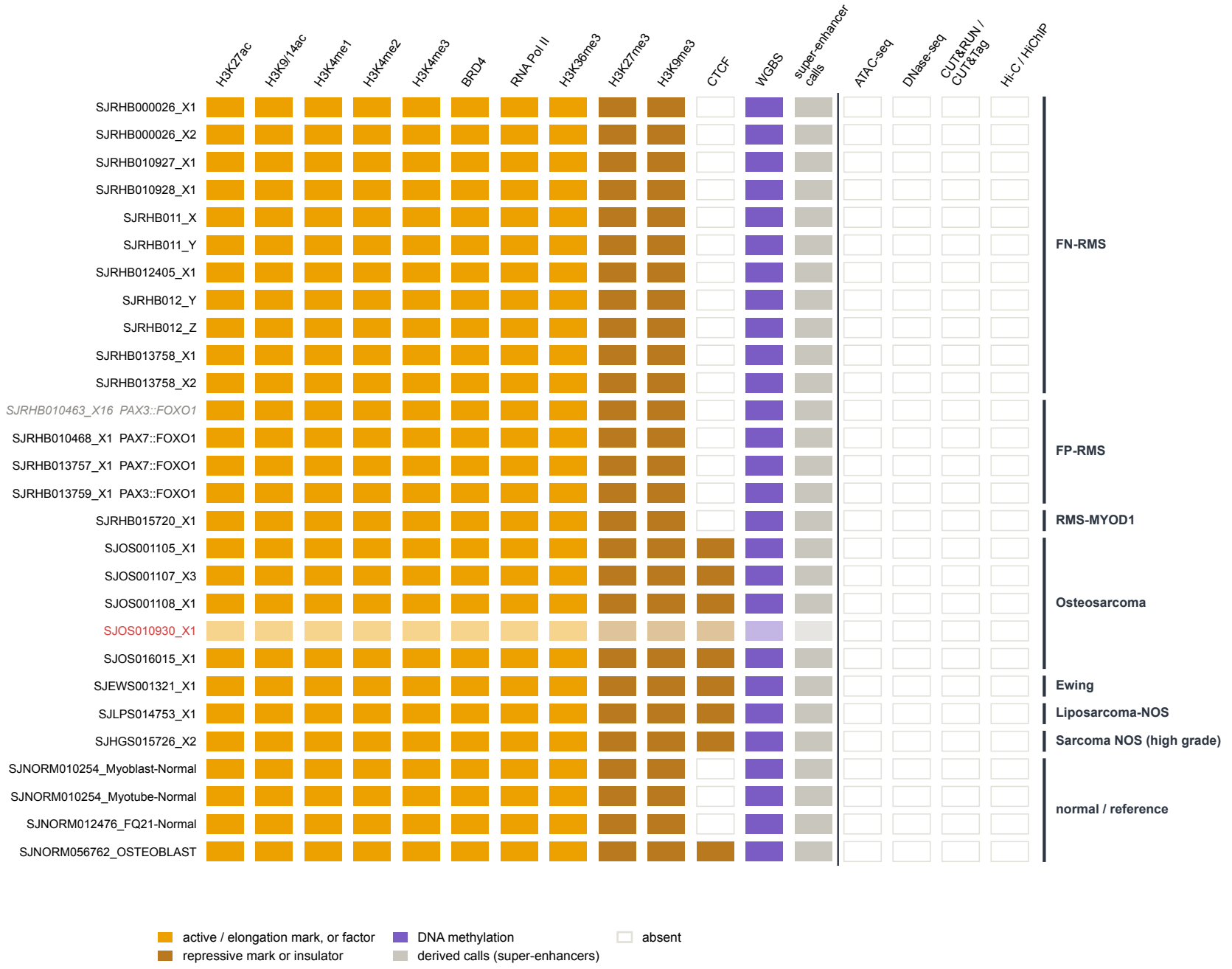
(d) The GDC has no ChIP-seq in its vocabulary either, and its 410 ATAC-seq files belong to 23 TCGA cohorts, none of them SARC. Methylation-array files exist for TARGET and TCGA sarcoma projects but are not sample-resolved here, so they are not counted.

(e) St Jude's COMET methylation project and the St Jude Cloud CSTN dataset. A request route exists for both, so they sit behind the same wall as EGA and dbGaP — but neither has an archive accession, which makes them the weakest case within it.

Also recovered this round: the CSTN ProteinPaint browsers resolve to 278 regulatory tracks on 27 models (T28, F26), with the per-model diagnosis supplied by the St Jude data administrator. That material is already counted once inside the EGA bar, so it is not added again — what it adds is assay-level resolution EGA does not publish.

The densest sarcoma epigenome resource that exists is complete, uniform — and entirely behind a wall.

Every one of the 28 St Jude CSTN models carries the same 10-mark panel plus WGBS and super-enhancer calls (CTCF only in the 2018 study). Right of the rule: what it does not contain at all.



Access: all of it requires a St Jude / EGA data access agreement. None of it is added to any atlas total — the same material is already counted once under EGA (F25). What the enumeration adds is this resolution, which EGA does not publish.

Entity calls are the St Jude data administrator's (email); the fusion partner is from the CSTN portal model table (T17). Both are recorded per track in T28.

Red label: SJOS010930_X1, which the administrator reports as withdrawn from CSTN and no longer available — its tracks are still drawn on the browser page and are shown faded.

Grey italic label: SJRHB010463_X16, which is on the browser page and in the portal as PAX3::FOXO1 but is absent from the administrator's list; the discrepancy is left standing.